Assignment 3: Github

Harsha Helsing, Manya Gupta

2026-11-09

Assignment 3: Collaborating in Github

Introduction

This is the base R markdown file we can populate with our code via branching, then we can replace this section with the introduction and data description, as well as the planned plots/analyses.

Installing and Loading Packages

```
#Feel free to add any and all packages we might need
#install.packages("pastecs")
library(pastecs)
#install.packages("tidyverse")
library(tidyverse)
#install.packages("knitr")
library(knitr)
#install.packages("car")
library(car)
```

Data Preparation

```
#Reading in csv file and assigning to data frame object named "text_wide"
text_wide <- read.csv("TextMessages.csv")
class(text_wide$Group)
```

```
## [1] "integer"
```

```
text_wide$Group <-factor(text_wide$Group,levels=c(1,2))

#In case we need long format data-
text_long <- pivot_longer(
  text_wide,
  cols= c("Baseline","Six_months"),
  names_to = "Timepoint",
  values_to = "Text_count"
  )
class(text_long$Timepoint)
```

```
## [1] "character"
```

```
text_long$Timepoint<-factor(text_long$Timepoint)

#Moving participant number to the front
```

```
text_wide <- text_wide |> relocate("Participant")
text_long <- text_long |> relocate("Participant")
```

Visualization 1: Faceted Boxplot of text messages by Group and Time

```
#Code chunk for boxplot
```

Visualization 2: Faceted Bar Chart of text messages by Group and Time

```
#Code chunk for bar chart
```

Summary Statistics of Text Messages by Group and Time

```
#We are first grouping the data by the group number and time-point, then
#computing the summary statistics that we want such as the mean, median, SD,
#range, etc., using dplyr from tidyverse.

sum_stat<- text_long |>
  group_by(Group,Timepoint) |>
  summarise(
  N=n(),
  Mean=mean(Text_count),
  SD=sd(Text_count),
  Median=median(Text_count),
  Min=min(Text_count),
  Max=max(Text_count),
  IQR=IQR(Text_count),
  .groups = "drop_last"
)

#To output a clean table of the summary statistics, we are using the kable
#function.

kable(sum_stat,
  caption="Summary Statistics of Text Messages by Group and Time-point",
  digits=2,
  align=c("c","l","c","c","c","c","c","c","c")
)
```

Table 1: Summary Statistics of Text Messages by Group and Time-point

Group	Timepoint	N	Mean	SD	Median	Min	Max	IQR
1	Baseline	25	64.84	10.68	64	47	85	16
1	Six_months	25	52.96	16.33	58	9	78	11
2	Baseline	25	65.60	10.84	65	46	89	14
2	Six_months	25	61.84	9.41	62	46	79	8

```
#At baseline, the two groups had relatively similar mean numbers of text
#messages. Group 1 had a mean of 64.84 (SD = 10.68), while Group 2 had a
#slightly higher mean of 65.60 (SD = 10.84). The similar means and standard
```

```
#deviations suggest that the groups had comparable text messaging patterns at
#the start of the study.

#At six months, both groups experienced a decrease in the mean number of text
#messages. However, the magnitude of the decrease differed between groups. Group
#1 had a mean of 52.96 (SD = 16.33), representing an average decrease of 11.88
#messages from baseline. Group 2 had a smaller decrease, with a mean of 61.84
#(SD = 9.41), representing a reduction of 3.76 messages.

#When comparing means to medians, most of the subgroups have similar values
#hinting at symmetrical distributions. We only see a major difference in the six
#month time-point for group 1 where the median was higher than the mean,
#suggesting a left skew to the distribution. There were also differences in the
#variability of text messaging patterns at six months. Group 1 demonstrated
#greater variability (SD = 16.33) compared to baseline, suggesting that
#participants' messaging behaviors became more dispersed. This is also reflected
#in the wider range of values (9-78). In contrast, Group 2 demonstrated less
#variability at six months (SD = 9.41), with a narrower range (46-79),
#suggesting more consistent messaging patterns among participants.
```

Interpretation and conclusions

Any concluding comments in addition to individual interpretations above